Solving quadratic equations by factor

1

a $x=0, x=-6$

b $m=0, m=-2$

c $x=0, x=\frac{12}{7}$

d $x=7, x=6$

e $x=7, x=-8$

f $x=\frac{19}{17}, x=\frac{5}{11}$

g $x=-\frac{19}{7}, x=\frac{5}{2}$

h $x=\frac{17}{4}, x=-\frac{11}{3}$

i $x=\frac{9}{4}$

j $x=1$

k $m=0, m=10$

2

a $x=0, x=-12$

b $y=0, y=\frac{1}{5}$

c $x=0, x=-4$

d $x=-10, x=-9$

e $x=-11, x=5$

f $x=9, 8$

g $x=10$

h $x=7, x=-2$

i $x=2, x=-4$

j $x=-4, x=5$

k $x=-\frac{2}{5}, x=-4$

l $x=\frac{9}{4}, x=4$

m $x=\frac{5}{4}, x=3$

n $x=-\frac{5}{4}, x=-3$

o $x=-\frac{4}{5}, x=-2$

p $x=-\frac{5}{3}, x=4$

q $x=-8, x=8$

r $x=\frac{8}{9}, x=\frac{-8}{9}$

3

a $x=0, x=32$

b 32 cm

4

a $m=14, 13$

b $m=5, m=-4$

c $m=0, m=14$

d $m=0, m=\frac{7}{11}$

e $y=-1, y=3$

f $x=\frac{-5}{2}, x=12$

5

a -10

b 1

7 $x=2, x=-3$

8

a $t=4, t=8$

b $4, 4$

9 $k=-16$

10 $a=13$

11

a No

b No

c No

d Yes

Additional questions

12

a

i $(0,\ 6)$

ii $(-1,\ 0)\,,\ (6,\ 0)$

iii $y=-\left(x-6\right)\left(x+1\right)$

b

i $(0,\ 12)$

ii $(2,\ 0)\,,\ (6,\ 0)$

iii $y=\left(x-2\right)\left(x-6\right)$

c

i $(0,\ -16)$

ii $(4,\ 0)$

iii $y=-\left(x-4\right)^2$

d

i $(0,\ 0)$

ii $(-5,\ 0)\,,\ (0,\ 0)$

iii $y=x\left(x+5\right)$

13 10 yards